

Title: Weather Dashboard Web Application

Course: DLBCSPJWD01 – Project Java and Web
Development

Tutor: Christian Remfert

Student: Muhammad Saad Bin Khurram

Matriculation Number:10220341

[GitHub Repository](#)

Purpose of the Application

Real-time weather updates for any city using the OpenWeatherMap API.



Quick city search through a simple and intuitive search bar.



Save favorite cities for faster access to frequent locations.



Responsive design for seamless use on desktop, tablet, or mobile.



Target Users & Benefits ↗

Target Users:

- Students & employees (to plan daily routines).
- Travelers (to check destination weather).
- General users who want quick updates.
- Benefits:
 - Saves time by providing instant forecasts.
 - Favorites list gives quick access to frequently checked locations.
 - Responsive design ensures usability on desktop, tablet, and mobile.

Technical Design Choices

Frontend: HTML, CSS, JavaScript → Responsive UI.

Backend: Node.js (Express) → Handles API requests.

Database: PostgreSQL → Stores favorite cities.

API Integration: OpenWeatherMap (free API).

Dynamic Aspects:

1. Search city → fetch & display weather.
2. Save/remove cities in “Favorites” list.

Summary & Next Steps

The Weather Dashboard is a simple and user-friendly web application that helps people check real-time weather updates for any city around the world. It combines a clean, responsive front-end design with a reliable Node.js back-end connected to the OpenWeatherMap API.

With just a quick search, users can view essential weather details like temperature, humidity, and wind speed, along with easy-to-understand weather icons. The app also allows users to save their favorite cities in a personal list, making it effortless to check the weather in those places again later.

Overall, the project focuses on making weather information fast, clear, and accessible from any device whether you're at home, at work, or on the go. In the next phase, I will begin implementing the core features and connecting the app to the weather API



Phase II

Technology	What it does in your app	Why you chose it
HTML / CSS / JavaScript	Frontend UI – search, weather card, favorites list	Required by the course; standard web technologies
Node.js	Runs the backend server	JavaScript on both frontend and backend; simple, lightweight
Express.js	Handles HTTP routes (/weather/:city, /favorites)	Minimal framework, easy to set up REST API endpoints
PostgreSQL	Stores favorite cities persistently	Relational database; reliable, free, supports conflict handling (ON CONFLICT)
Axios	Backend makes HTTP requests to OpenWeatherMap	Clean promise-based API client; better error handling than native fetch in Node
dotenv	Loads .env config (API key, DB credentials)	Keeps secrets out of source code
OpenWeatherMap API	Provides real-time weather data	Free tier available; returns JSON with temp, humidity, wind, icons
cors	Allows frontend to call backend during development	Required when frontend and backend run on different ports

Changes from Phase 1

Area	Phase 1 Plan	What Actually Changed
Weather data displayed	Temp, humidity, wind speed, icons	Added: feels like, temp min/max, weather description, animated weather icons
UI / theming	Basic responsive layout	Dynamic card theming based on weather condition (e.g., blue for rain, yellow for clear) + loading spinner animation
City search	Simple text input → search	Added autocomplete suggestions from a cities.json list as user types
Favorites	Save/remove cities	Fully implemented with PostgreSQL – persists between sessions, sorted alphabetically
API security	OpenWeatherMap consumed by backend	Implemented correctly – API key never exposed to the frontend

#	Feature	Input	Expected Result	Actual Result	Pass/Fail
1	City search – valid	Type "Berlin", click Search	Weather card shows temp, humidity, wind for Berlin	Card displays Berlin weather with correct data	✓ Pass
2	City search – invalid	Type "xyzabc123", click Search	Error message: "Could not fetch weather for that city."	Error message displayed	✓ Pass
3	City search – empty	Click Search with empty input	Message: "Please enter a city."	Prompt shown, no API call made	✓ Pass
4	Autocomplete	Type "Ber" in input	Dropdown shows cities starting with "Ber" (Berlin, Bergen...)	Suggestions appear, click selects city and searches	✓ Pass
5	Add to Favorites	Search "London", click ★ Add to Favorites	"London" appears in Favorites list	London added and shown in sidebar	✓ Pass
6	Duplicate favorite	Add "London" twice	City added only once (no duplicates)	PostgreSQL ON CONFLICT DO NOTHING prevents duplicate	✓ Pass
7	Remove from Favorites	Click × next to "London"	London removed from list	Deleted from DB, list updates immediately	✓ Pass
8	Click favorite city	Click 🔍 Berlin in Favorites	Weather card updates to Berlin	Weather fetched and displayed	✓ Pass
9	Responsive layout	Resize browser to mobile width	Layout stacks vertically, input/buttons usable	Layout adjusts correctly on small screens	✓ Pass
10	Persistence	Add "Paris", restart server, reload page	Paris still in Favorites	Data persists in PostgreSQL between sessions	✓ Pass

#	Feature	Input	Expected Result	Actual Result	Pass/Fail
11	Dynamic card theming	Search a rainy city (e.g. "Bergen")	Weather card changes color to rain theme	Blue accent applied to card	✓ Pass
12	Loading spinner	Click Search	Spinner shows while data loads	Spinner appears, disappears when data arrives	✓ Pass
13	Responsive layout – mobile	Resize browser to 375px width	Layout stacks vertically, all elements usable	Layout adjusts correctly	✓ Pass
14	Responsive layout – tablet	Resize browser to 768px width	Layout adapts to medium screen	Elements scale and reposition correctly	✓ Pass
15	API key hidden from frontend	Open browser DevTools → Network tab	API key not visible in any frontend request	All API calls go through backend only	✓ Pass



Screenshots and Screencast

Cities search

Button to search cities

Weather Dashboard

Search for a city to see the weather.

Favorites

 Berlin 

 Karachi 

Results

Cities added to favorite

Weather Dashboard

Berlin

Search

Favorites

Berlin, DE


few clouds

17°C

Feels like 17°C

HUMIDITY

 55%

WIND

 10.28 m/s

RANGE

↓16° ↑18°

★ Add to Favorites

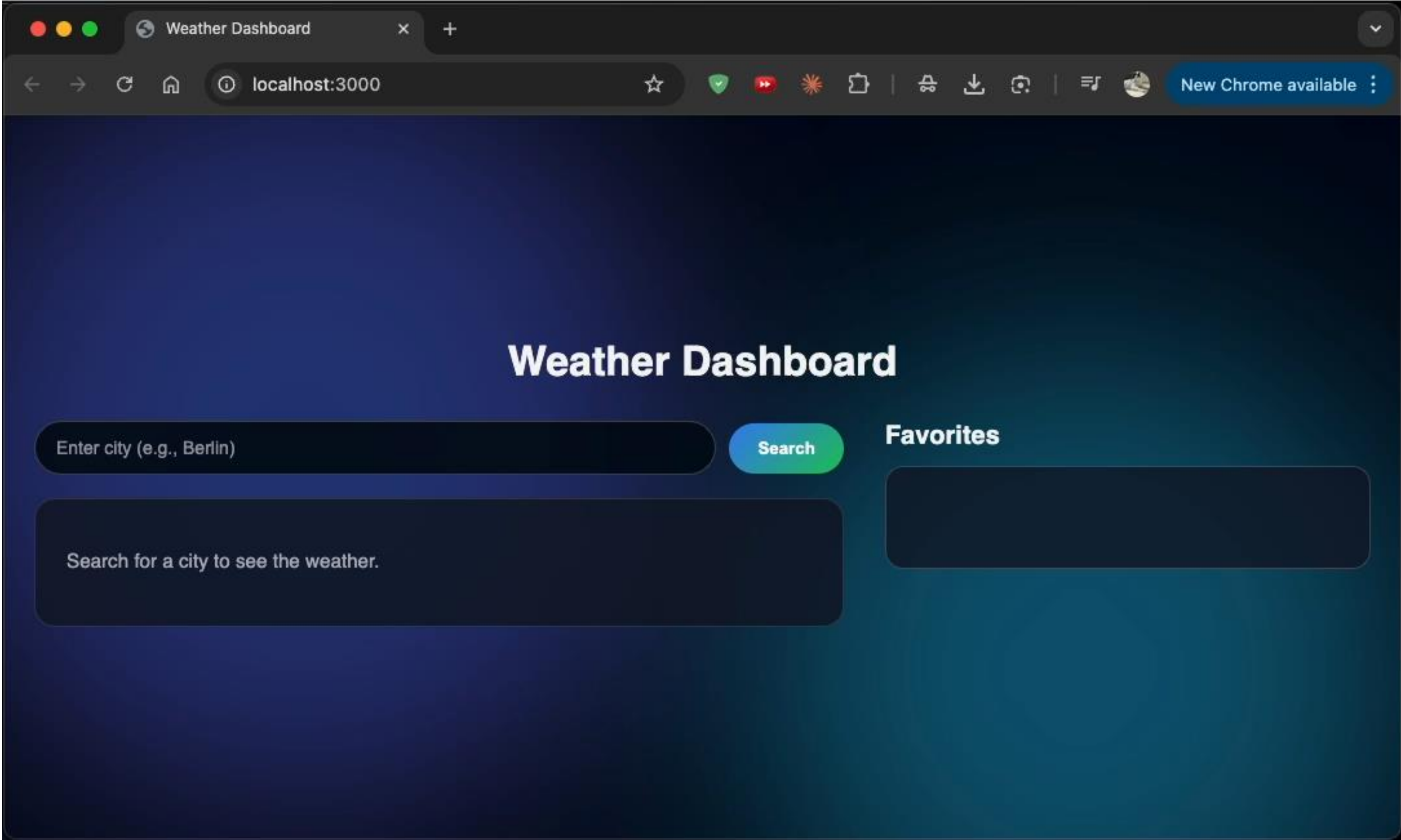
 Berlin ✕

 Karachi ✕

Results

Mobile View





Conclusion

The Weather Dashboard is a full-stack web application developed as part of the IU course DLBCSPJWD01. The application allows users to search for real-time weather data for any city in the world, view detailed weather information, and manage a personal list of favourite cities that persists across sessions. The frontend is built with HTML, CSS, and JavaScript, featuring a responsive layout that adapts to desktop, tablet, and mobile screens. The backend is powered by Node.js and Express, which handles all API communication with OpenWeatherMap and manages favourite cities through a PostgreSQL database.

Throughout the development phase, several enhancements were made beyond the original Phase 1 concept — including dynamic weather-based card theming, a city autocomplete search feature, animated loading states, and a more detailed weather display showing feels-like temperature, daily min/max range, and wind speed. All core features were tested and verified, covering valid and invalid inputs, data persistence, responsive behavior, and API security.

The application meets all technical requirements set by the course: at least two dynamic JavaScript interactions with the backend, responsive design, external API consumed exclusively by the backend, and full code documentation.

THANKYOU

